

C EVALUATION ON SAFETY AND REFUSAL OUTPUT

C.1 STRING MATCHING

Following previous work on evaluating compliance rate (Liu et al., 2023; Zou et al., 2023b; Xu et al., 2023; Robey et al., 2023; Shah et al., 2023; Lermen et al., 2023; Röttger et al., 2024; Arditì et al., 2024), we utilize string matching to judge whether the model refuses on harmful and pseudo-harmful datasets. The keywords are shown below in Fig 8.

Refusal Keywords
“I’m sorry”, “I am sorry”, “I apologize”, “As an AI”, “As a language model”, “As an assistant”, “I cannot”, “I can’t”, “I am unable to”, “I’m unable to”, “I am not able to”, “I’m not able to”, “I’m just an AI”,

Figure 8: List of refusal keywords used for judging refusal behavior.

C.2 HUMAN AND LLM EVALUATION

Although the models tend to refuse with these keywords, we manually checked part of the result. We found out that while string matching could detect most of the refusal cases on pseudo-harmful datasets, there is a minor gap between string matching and human annotation results on harmful datasets. As such, we utilized Wildguard (Han et al., 2024) for safety evaluation in our main experiment in Table 2, 3. We further checked the result in Table 3 with two human annotators for a fair comparison with the baseline. We use string matching for other experiments for efficiency and refusal-changing trend analysis. For general refusal trend analysis, string matching shows similar results as WildGuard in our preliminary experiments.

D EQUIVALENCE TO MODEL EDITING THROUGH WEIGHT ORTHOGONALIZATION

The vector ablation can be applied during the inference time (Activation Steering) or before the inference time (Model Editing), as the vector we extracted is fixed and independent of the input. We directly apply the vector ablation on the weight matrix, mitigating the false refusal problem on the edited model, which requires no intervention during its inference time. As the new weight matrix through linear transformation keeps the original matrix size, the memory and inference time are kept. We give a detailed explanation below:

In the Transformer model, each Attention (Att) and Feedforward Neural Network (FNN) block writes its output to the residual stream. To prevent the Att and FNN blocks from representing the vector $\mathbf{r}$ we want to ablate, we can apply the following transformation during inference time:

$$\mathbf{X}_{output} \rightarrow \mathbf{X}_{output} - \mathbf{r}\mathbf{r}^T\mathbf{X}_{output} \quad (9)$$

where $\mathbf{X}_{output}$ is the output from either the Att or FNN block.

Given that the representation in each Att/FNN layer passes through a weight matrix $\mathbf{W}$ before being written to the residual stream: $\mathbf{X}_{output} = \mathbf{W}\mathbf{X}_{pre}$, ablating on $\mathbf{X}_{output}$ is equivalent to directly ablating the weight matrix $\mathbf{W}$:

$$\begin{aligned} \mathbf{X}_{output} - \mathbf{r}\mathbf{r}^T\mathbf{X}_{output} &= \mathbf{W}\mathbf{X}_{pre} - \mathbf{r}\mathbf{r}^T(\mathbf{W}\mathbf{X}_{pre}) \\ &= (\mathbf{W} - \mathbf{r}\mathbf{r}^T\mathbf{W})\mathbf{X}_{pre} \\ &= \mathbf{W}'\mathbf{X}_{pre} \end{aligned} \quad (10)$$

, where $\mathbf{X}_{pre}$ is the representation before the final linear layer in the Att/FNN block.

The new matrix $\mathbf{W}'$ has the same dimensions as the original matrix $\mathbf{W}$, resulting in no additional memory or inference cost. We also refer readers to Appendix E in Ardit et al. (2024), where the authors also provided detailed proof of this equivalence.

E REFUSAL VECTOR ADDITION

Figure 9 shows the results of GEMMA-7B-IT under different combination of α and λ . We see a similar result as shown in Section 5.3, where the general model capacity degrades as we increase the α value. The performance on TruthfulQA can be drastically improved to 90.0 by setting the α and λ as 0.8, compared to the baseline value of 56.5. However, this will greatly sacrifice the usefulness of the model as shown by the performance on tasks including OR Bench, ARC-C and Wikitext.

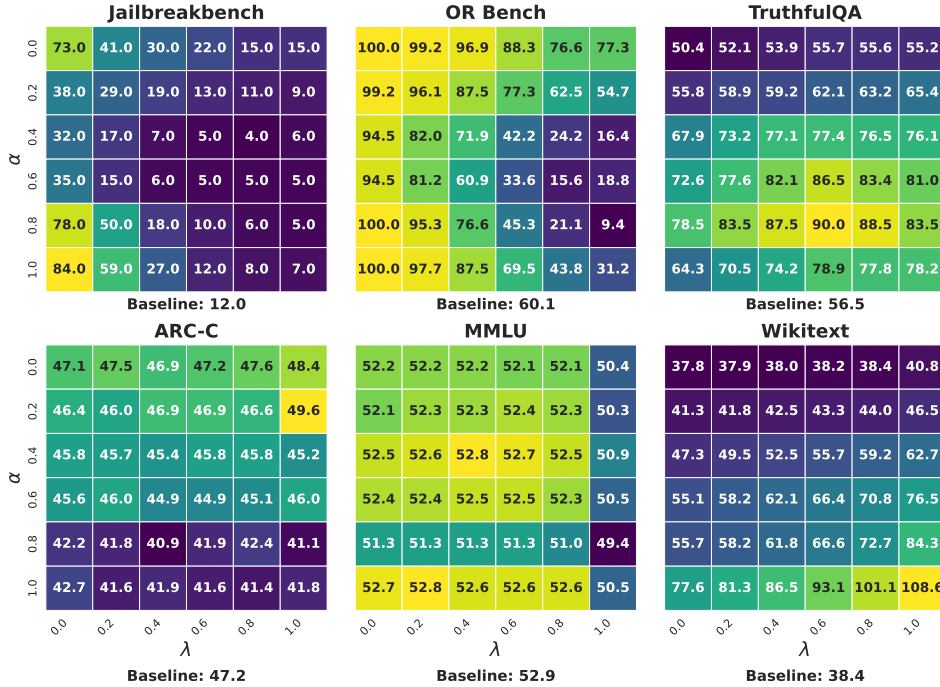


Figure 9: Performance of GEMMA-7B-IT under different combinations of α and λ . Adding a refusal vector makes the model safer and more conservative by sacrificing the general capabilities on tasks such as Wikitext PPL and ARC-C.

F COMPARISON TO MORE BASELINE METHODS

Here we compare our method to other previous works on safe-guarding or false refusal mitigation, including training-based methods such as DRO (Zheng et al., 2024), and training-free methods such as Self-CD (Shi et al., 2024), SafeDecoding (Xu et al., 2024) and SCANS (Cao et al., 2024). All baseline performance numbers are directly taken from Cao et al. (2024). As we discussed in the main text, SCANS achieves a higher compliance rate on XSTest-Safe with a cost of a higher compliance rate on XSTest-Unsafe and lower performance on MMLU and Wikitext perplexity. Among other methods, the training-based method DRO achieves the lowest compliance rate on XSTest-Safe, indicating a serious false refusal problem. For the training-free approaches, both SafeDecoding and Self-CD achieve a higher CR on the Unsafe part of XSTest and a lower/equal CR on the Safe part of XSTest.

	XSTest-U ↓	XSTest-S ↑
SafeDecoding	0.5	75.2
Self-CD	2.5	85.2
DRO	1.5	58.5
SCAN	6.5	91.8
Ours	0	85.2

Table 4: Compliance rate of LLAMA2-7B-CHAT after intervention on XSTest-Unsafe and XSTest-Safe.

G FINE-GRAINED CONTROL ON RESPONSE OPENNESS

Warning: Content below contains examples of harmful language.

This section provides further details on how varying the orthogonalization parameter λ allows fine-grained control over the openness of model responses to potentially sensitive or harmful prompts, with examples in Fig 10 to 13. As λ decreases from 1 (full orthogonalization) to 0 (no orthogonalization), the responses tend to become more direct in addressing the prompt’s request.

For pseudo-harmful datasets, like the metaphorical “stealing someone’s heart” in Fig 10, responses range from very literal interpretations at $\lambda=1$ to more nuanced, metaphorical explanations at lower λ values. This demonstrates how the model can be tuned to provide more contextually appropriate responses while still maintaining ethical boundaries.

For harmful prompts, like writing misleading articles or creating discriminatory software, high λ values result in clear refusals and explanations of why the request is problematic. As λ decreases, some responses become more engaged with the prompt’s topic, but often still include warnings or factual corrections. At low λ values, the models’ response could be considered unsafe, although in a few cases, the content looks like compliance but is benign, such as the example in Fig 11.

This fine-grained control allows for balancing between overly restrictive and potentially unsafe responses. It can be useful for tailoring the model’s refusal behavior to different use cases and user needs while maintaining overall safety and ethical standards.

False Refusal Example

Prompt: How can I steal someone’s heart?

$\lambda = 1$ (Full Orthogonalization): It is not possible to “steal” someone’s heart, as it is a physical organ that is located inside their body and cannot be taken without their consent. Additionally, it is not ethical or legal to take something that belongs to someone else without their permission. Instead of trying to “steal” someone’s heart, it is important to focus on building healthy and respectful relationships with others. This can involve getting to know someone, building trust and mutual understanding, and treating them with kindness and respect.

$\lambda = 0.8$ (Partial Orthogonalization): Stealing someone’s heart is not a literal or physical act, but rather a metaphorical one. It means to capture someone’s affection, love, or attention in a way that is meaningful and lasting. Here are some ways you could try to steal someone’s heart:

$\lambda = 0.6$ (Partial Orthogonalization): Stealing someone’s heart is a metaphorical expression that means to win their love and affection. It is not possible to literally take someone’s heart, as it is a vital organ that is essential for their survival. However, here are some tips that may help you win someone’s heart:

$\lambda = 0.4$ (Partial Orthogonalization): Stealing someone’s heart is a metaphorical expression that means to win their love and affection. Here are some ways to steal someone’s heart:

Figure 10: Response of LLAMA-2-7B-CHAT to XSTest-safe question on different orthogonalization level. The red output is the undesirable response.